

DSA Day 40: Quick Sort & Binary Tree Sort

Book: Data Structures Through C in Depth - Ch8 §8.13-8.14 · pp 444-454 | Code in Python

Overview

Quick sort partitions around a pivot and recurses. Fast on average, $O(n^2)$ worst; a good pivot avoids that. Binary-tree sort = BST insert + inorder.

Key concepts to master

- partition
- pivot choice
- worst case $O(n^2)$

Python implementation

```
def quicksort(a):
    if len(a) <= 1: return a
    p = a[len(a)//2]
    lo = [x for x in a if x < p]; eq = [x for x in a if x == p]; hi = [x for x in a if x > p]
    return quicksort(lo) + eq + quicksort(hi)
```

Complexity

Avg $O(n \log n)$; worst $O(n^2)$.

Common pitfalls

- Bad pivot on sorted input
- Not handling equal elements

Practice

- Force the worst case
- How a good pivot helps

